# AdaDpFed: A Differentially Private Federated Learning Algorithm With Adaptive Noise on Non-IID Data

Zirun Zhao, Yi Sun, *Member, IEEE*, Ali Kashif Bashir, *Senior Member, IEEE*, and Zhaowen Lin

***Abstract*—The popularity of emerging consumer electronics, such as Mobile phones, PADs, and various smart home appliances, brings unprecedented convenience to people. Currently, how to obtain targeted information without revealing personal privacy is the new challenge with the application of precision recommendation and other Artificial Intelligence technologies in consumer electronics. Several solutions, based on Federated Learning transfer and update models by intermediate parameters instead of individual data, can perform some recommending tasks with weak security. However, they still face various attacks like model inversion attacks and membership inference attacks, especially in the complex non-independent and identically distribution (non-IID) environment. In this paper, we propose AdaDpFed, an adaptive federated differential private protocol in the non-IID setting. AdaDpFed can adaptively adjust the perturbation parameters, aggregated clients and sampling size according to the changing distribution of the individual data. The convergence proof shows that AdaDpFed has $\mathcal{O}(\frac{1}{T})$ convergence rate. Comparative experiments demonstrate that the performance of AdaDpFed outperforms other state-of-the-art protocols in both accuracy and global privacy budget.**

***Index Terms*—Personalized recommendation system, federated learning, differential privacy, adaptability, quantification.**

## I. Introduction

THE WIDESPREAD popularity of consumer electronics has spawned plenty of intelligent applications such as smart assistants, social platforms, and various online shopping software. To provide better customer service, almost all intelligent applications are collecting individual data for accurate personalized recommendations. However, this may raise concerns about users' private information being compromised. Some solutions, based on federated learning, are replacing the transfer of original data with model updates to ensure the private data is preserved in local devices.

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Unfortunately, attacks [1], [2] like model inversion attack or membership inference attack can still reveal the personal data from the updates of the models. Techniques using differential privacy (DP) [3] have been proposed to address this issue, but also bring challenges. First, the introduction of DP would inevitably reduce the accuracy of the model. Second, most DP-based federated learning protocols ignore the potential existence of malicious clients and adopt unified budget allocation which is almost useless for defense. The situation may be even worse in the non-IID setting. Therefore, the way achieving higher accuracy with adaptive privacy budgets becomes the key problem in real-world federated learning.

Currently, DPSGD [4] is the most widely used method to perturb the gradients during the model training. It uses moments accountant to track the accumulated budgets. Many applications in natural language processing [5], computer vision [6] and healthcare analysis [7] have adopted DPSGD to protect data on remote devices.

Based on DPSGD, Yang and Dong [8] proposed a method that reduces the impact of the noise by Kalman filtering but adopted uniform budget allocation and ignored the dynamic change of distinction. Zhao et al. [9] analyzed the accuracy based on 1-Wasserstein distance and designed a novel adaptive differentially private federated learning algorithm (A-DPFL). Xiong et al. [10] proposed a 2DP-FL to achieve a flexible noise addition but introduced two rounds of noise superimposition. These studies integrate differential privacy with federated learning but lack specific budget allocation methods combined with real-world considerations, such as non-IID setting, stragglers and sample redundancy. Therefore, we provide algorithms based on the degree of client drift and the dynamic data distribution to train local models with adaptive budget. The specific contributions are listed as follows.

- We propose a quantification mechanism to precisely evaluate the impacts by introducing DP and measure the degree of non-IID. With the calculation of the quantification values in each round, an aggregation method is designed to filter the top $K$ results that meet the predetermined thresholds, which improves the time efficiency when ensuring the accuracy of the aggregated model.
- We propose an adaptive federated differential private protocol, AdaDpFed, based on the quantification mechanism in non-IID. It uses quantification values to evaluate the non-IID degree of clients in each round of aggregation and perturb the returned gradients adaptively based on

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the degree to reduce the impact of non-IID on the convergence speed.
- We adopt an adaptive sampling method based on a sliding window to increase the probability of learning new features and accelerate model optimization. It makes AdaDpFed more suitable for real random industrialized IoT scenarios.
- We prove that AdaDpFed has $\mathcal{O}(\frac{1}{T})$ convergence rate. The experiments also show that the proposed AdaDpFed outperforms FedAvg+DPSGD in terms of convergence speed and accuracy.

## II. Preliminaries

### A. Differential Privacy

DP defines a parameterized mathematical model that ensures any two neighboring databases can get the same output with almost the same probability. It preserves the privacy of individuals without influencing the statistical properties of the database by adding noise with appropriate scales. DP is formally defined as follows:

*Definition 1:* A randomized algorithm $\mathcal{M}$ with domain $\mathbb{N}^{|\mathcal{X}|}$ sacrifices $(\epsilon, \delta)$-**differential privacy** if for all $O \subseteq Range(\mathcal{M})$ and neighbouring inputs $d, d' \in \mathbb{N}^{|\mathcal{X}|}$:

$$\Pr[\mathcal{M}(d) \in O] \leq e^{\epsilon} \cdot \Pr\left[\mathcal{M}\left(d'\right) \in O\right] + \delta \tag{1}$$

When $\delta = 0$, we say it equals to $\epsilon$-differential privacy. Dwork et al. [11] shows that $(\epsilon, \delta)$-differential privacy is a relaxation of the plain $\epsilon$-differential privacy with the broken probability $\delta$. Therefore, with the variance calibrated to $\frac{\Delta f \cdot \ln(1/\delta)}{\epsilon}$, one can achieve $(\epsilon, \delta)$-differential privacy.

Two basic composition theorems [12] are given to describe the composability and post-processing of DP.

*Theorem 1:* Let $\mathcal{M}_i : \mathbb{N}^{|\mathcal{X}|} \to R^d$ be an $(\epsilon_i, \delta_i)$-differential private algorithm for $i \in [k]$. Then if $\mathcal{M}_{[k]} : \mathbb{N}^{|\mathcal{X}|} \to \prod_{i=1}^{k} R_i^d$ is defined to be $\mathcal{M}_{[k]}(x) = (\mathcal{M}_1(x), \ldots, \mathcal{M}_k(x))$, then $\mathcal{M}_{[k]}$ is $(\sum_{i=1}^{k} \epsilon_i, \sum_{i=1}^{k} \delta_i)$-differential private under **basic composition theorem**.

*Theorem 2:* For all $\epsilon, \delta, \delta' \geq 0$, the class of $(\epsilon, \delta)$-differentially private mechanisms satisfies $(\epsilon', k\delta + \delta')$-differential privacy under $k$-**fold adaptive composition** for:

$$\epsilon' = \sqrt{2k\ln(1/\delta')}\epsilon + k\epsilon\left(e^{\epsilon} - 1\right) \tag{2}$$

An immediate and useful corollary tells a safe choice of $\epsilon$ for each of $k$ mechanisms if we wish to ensure $(\epsilon', k\delta + \delta')$-differential privacy for a given $\epsilon', \delta'$.

*Corollary 1:* Given target privacy parameters $0 < \epsilon' < 1$ and $\delta' > 0$, to ensure $(\epsilon', k\delta + \delta')$ cumulative privacy loss over $k$ mechanisms, it suffices that each mechanism is $(\epsilon, \delta)$-differentially private, where $\epsilon = \frac{\epsilon'}{2\sqrt{2k\ln(1/\delta')}}$.

Abadi et al. [4] reduce the upper bound further by moments accountant in DPSGD.

*Theorem 3:* There exist constants $c_1$ and $c_2$ so that given the sampling probability $q = L/N$ and the number of steps $T$, for any $\epsilon < c_1 q^2 T$, DPSGD is $(\epsilon, \delta)$-differentially private for any $\delta > 0$ if choose $\sigma \geq c_2 \frac{q\sqrt{T\ln(1/\delta)}}{\epsilon}$.

### B. Federated Learning

A federated learning system generally consists of an aggregating server and $n$ local clients. Each client $i \in [n]$ collects a local database $\mathcal{D}_i$ with $n_i$ samples. The most widely used algorithm FedAvg [13] is shown as follows.

*FedAvg:* The purpose of FedAvg is to find the optimal $\mathbf{w}$ to get the minimum value of the global loss function. In FedAvg, the server is responsible for aggregation and the data are trained by local devices. The optimization problem is described as follows:

$$\texttt{FedAvg}: \min_{\mathbf{w} \in \mathbb{R}^d} F(\mathbf{w}) \stackrel{\text{def}}{=} \sum_{k=1}^{K} p_k F_k(\mathbf{w}) \tag{3}$$

where $F_k(\mathbf{w}) = \frac{1}{n_k} \sum_{i \in \mathcal{D}_k} F_i(\mathbf{w}), p_k = \frac{n_k}{n}, n_k = |\mathcal{D}_k|, n = \sum_{k=1}^{K} |\mathcal{D}_k|$, $F_i(\mathbf{w})$ represents the local loss function $\ell(x_i; y_i; \mathbf{w})$.

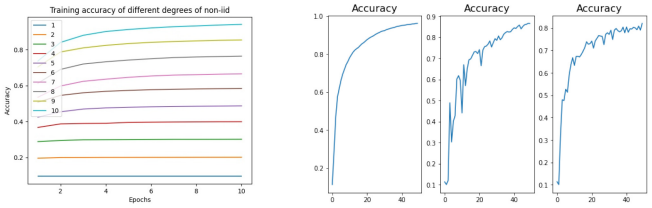
## III. Related Works

DP was proposed by Dwork to provide privacy for statistics publication with strict mathematical guarantees [3]. It has been widely applied to privacy preservation of areas like data publishing [14], federated learning [4] and edge intelligence [15].

Some research is working on utilizing relaxation to explore tighter bounds. Mironov [16] proposed Renyi-DP which applies to one moment at a time. Associating Renyi-DP with SGD optimizer, Abadi et al. [4] designed DPSGD with moments accountant to track privacy loss.

In federated learning, FedAvg gives a standard paradigm that unites devices to jointly train a model but it ignores the problem of system heterogeneity and statistical heterogeneity. FedProx [17] is proposed to alleviate the problem of heterogeneity by adding a proximal term to the original loss function. FedAvgM [18] proposed a method for combining datasets with a constant range of identicalness. Reddi et al. [19] proposed federated versions of ADAGRAD, ADAM, and YOGI. FedBN [20] uses local batch normalization to reduce the feature shift. Some research attempts to alleviate the drift issues in specific tasks. Jiang et al. [21] proposed a harmonizing framework to address the feature heterogeneity triggered by diverse scanners or protocols. Xu and Huang [22] used two momentum buffers to track and coordinate updating directions of the server model and the local model to reduce the influence of non-IID. Some studies are looking for substitutes to address the change of statistical heterogeneity. Tan et al. [23] proposed FedProto to replace gradients with prototypes in aggregation and accelerated the convergence of the global model. Yue et al. [24] replaced gradients with sample-wise Jacobian matrices to make updated data more expressive and efficient. There are also other improvements in different components of the traditional framework. He et al. [25] focused on the phenomenon that knowledge distillation may mislead local models and proposed a class-wise adaptive self-distillation mechanism to soften the influence of distillation loss. Qu et al. [26] proposed an effective algorithm based on Sharpness Aware Minimization to increase the generality of local clients. Tang et al. [27] conducts federated learning

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(a) Degree of Non-IID (b) IID vs Non-IID vs DP

Fig. 1. The influence of non-IID and DP.

with a virtual homogeneous dataset crafted to rectify the data heterogeneity. Chen et al. [28] pointed out that skewed labels lead to heterogenous local models and proposed a Calibrated FAT approach to balance the classes.

In addition to optimizer enhancements, some other methods have been used to improve the federated learning convergence under non-IID in combination with DP. Wei et al. [29] added DP-based noise twice at local training and parameter aggregation. Andrew et al. [30] replaced the fixed clipping norm with the adaptive threshold, which results in the adjustment of the noise scale as well. Pan et al. [31] utilized layer perturbation to protect the weights of teacher models and defend against external attacks, but it still maintains the fixed budget. However, these works basically ignored the dynamic change of distinction and lack the exploration of relationship between noise and distribution.

Some other technologies are also used in the federated framework with non-IID data such as secret data sharing [32], secure aggregation [33] and decentralized architecture [34].

## IV. Problem Statement

Federated learning has two practical dilemmas, i.e., non-IID setting and privacy issues. Non-IID is usually caused by client drift, i.e., the phenomenon that the distribution of data across clients is heterogeneous and it mostly slows down convergence. The privacy issues might be caused by various attacks. The introduction of DP can be a feasible method to resist attacks [4] but it also slows speed of the convergence. Therefore, how to achieve higher accuracy with a certain level of privacy has become an urgent problem to be solved.

We set two groups of experiments to explore the influence of non-IID and DP. Experiments select the MNIST handwritten digit dataset as the source dataset and resynthesize datasets for IID and non-IID settings. The first group explores the influence of the degree of non-IID on accuracy. 50 clients are separated into 10 groups to represent 10 levels of non-IID. The specific classes for clients of the same level are different. All the clients share the same test dataset which includes all classes. In Fig. 1(a), we calculate the average accuracy for different degrees of non-IID. It is intuitive that higher accuracy can be obtained with a higher degree of non-IID which represents a more reasonable data distribution.

The second group adopts the federated architecture (50 clients) and conducted the same experiments simultaneously in the IID, non-IID, and non-IID with DP. In the IID setting, samples are randomly assigned to any client. In the non-IID setting, we randomly initialize the level of non-IID for each client and then extract the samples of the corresponding number of classes. The specific classes are still random and all clients share the same test dataset. The non-IID setting with DP replaces SGD with DPSGD and then fully adopts the non-IID setting. The other training parameters are completely consistent and the training result is obtained as Fig. 1(b). It can be seen that the curve of the IID rises stably and finally converges to 96.1%. The curve of non-IID shows obvious oscillation and the final accuracy only reaches 86.5%. However, in the non-IID setting with DP, the oscillation amplitude of the curve has been significantly weakened and the final accuracy declines as expected to 82.6%.

The results of two groups of experiments are in line with our expectations and imply two effects of DP on non-IID.

*Oscillation amplitude:* Due to the noise added to the aggregated updates, the non-IID setting with DP weakens the amplitude of the oscillation. This coincides with some methods of using DP to reduce overfitting.

*Accuracy:* The DP can be realized by superimposing noise sampled from a distribution of certain parameters. If a learning algorithm adopts the DP-based optimizer, the noise would be added to the average gradient of batches or lots. The introduction of DP would inevitably reduce the accuracy of the model. However, just like the composition theory exploring bounds over multiple mechanisms, we hope to explore the upper bound of the model accuracy in the non-IID setting for the given budget.

## V. Adaptive Differential Private Federated Learning

### A. Method Design

A basic consensus is that non-IID can significantly affect the speed of convergence. Although the addition of DP can alleviate the convergence difficulties caused by non-IID, it inevitably affects the accuracy due to the introduction of noise.

DPSGD [4] is the most widely used algorithm combining DP and deep learning, but it has less exploration of non-IID and federated learning. While applying DP with a federated framework in the non-IID setting, the uniform noise addition may threaten the privacy of the local clients. It raises the question of how to adaptively perturb local gradients to obtain the fastest global convergence rate.

Based on the above considerations, we propose the following methods to improve the global model accuracy in the non-IID setting with DP. The framework is shown in Fig. 2.

*Adaptive Budget Allocation:* To ensure adaptive and reasonable noise, we first need to evaluate the degree of non-IID for each client. Considering that the minimum value calculated by each local model varies distinctly, we take the performance distinction between the global model of each round and the local optimal model of each client to be the quantification value. The absolute value of the quantification value can reflect the distinction of data distributions between the single client and the federated system, i.e., the degree of non-IID.

The quantification values are recalculated after each round. In the first round, the initialization module requires all clients

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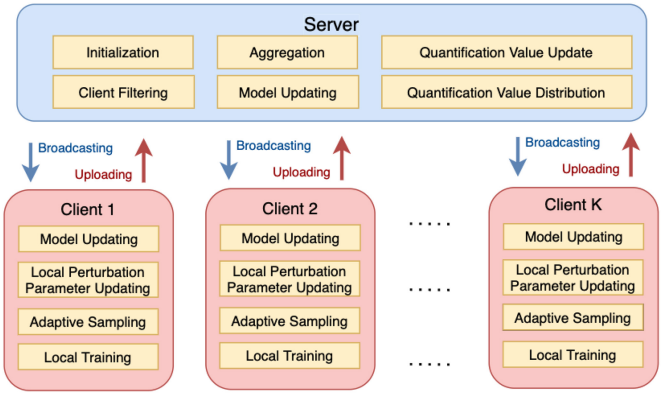



Fig. 2. The framework of AdaDpFed.

**Algorithm 1: AdaDpFed.** There Are $K$ Clients, $T$ Is the Total Number of Rounds, $S_t$ Is the Chosen Set of Round $t$, $d_t^i$ Is the Quantification Value for Client $i$ in Round $t$, $\Delta \mathbf{w}_t^i$ Is the Update From Client $i$ in Round $t$, $\mathbf{w}_t$ Is the Aggregated Model for Round $t$

**Init**($\mathbf{w}_{init}, K$)
**Server Executes**
**for** each round $t = 0, 1, ..., T$ **do**
  $S_t \leftarrow$ **Client Filtering**($K_n, Q, c$)
  **for** each client $i \in S_t$ in parallel:
    $\Delta \mathbf{w}_{t+1}^i \leftarrow$ **AdaptiveUpdate**($i, \mathbf{w}_t, d_t^i$)
    $\mathbf{w}_{t+1} \leftarrow \sum_{i=1}^{n} \frac{n_i}{n} \cdot \Delta \mathbf{w}_{t+1}^i + \mathbf{w}_t$
    if $t < T$:
      $d_{t+1}^i \leftarrow F(\mathbf{w}_{t+1}) - F(\mathbf{w}_t + \Delta \mathbf{w}_{t+1}^i)$
      *Clients* $\leftarrow (\mathbf{w}_{t+1}, d_{t+1}^i)$
return $\mathbf{w}_T$.

to participate to generate initial quantification values. In the subsequent rounds, the list of quantification values will be updated after client filtering and gradient aggregation because the sample pool of each client is assumed to be changing over time. It aims to simulate the collection of new samples and the diversity of non-IID. Then the updated quantification values are sent back to the corresponding clients to adaptively adjust the noise scale. The details are shown in Algorithm 1&2&3.

*Client Filtering:* In a real-world application, the federated framework is generally asynchronous. It is impractical to spend much time waiting for stragglers or unqualified updates caused by extreme distributions. For the above reasons, client filtering is proposed to select more reliable and powerful clients to accelerate the iterations. Depending on the quantification value and the preset threshold, clients can be quickly sorted and filtered by the server. The selection of the threshold can also balance the training efficiency and model performance. Details are shown in Algorithm 4.

*Adaptive Sliding Window.* In a real-world federated framework, client-side data collection is not a one-time event. The high-frequency data in IoT is generally collected from time-series sensors [35] which means the number of collected samples is growing over time. However, repeatedly learning old samples brings minor performance improvement but leads to huge calculating costs. Moreover, due to the fixed sampling

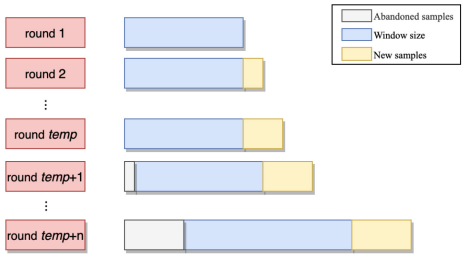



Fig. 3. Adaptive sliding window.

**Algorithm 2: Init.** $\mathbf{w}_{init}$ Is the Initial Parameter Vector Given by the Server. There Are $K$ Clients. $F(\cdot)$ Is the Global Objective Function, $F_i(\cdot)$ Is the Local Objective Defined by a User-Specified Loss Function $\ell(\cdot;\cdot)$, and $n_i$ Is the Number of Samples in Client $i \in K$

**Server Executes**
  Initialize $\mathbf{w}_{init}$
**Clients Executes**
  **for** each client $i \in K$ **do**:
    $\mathbf{w}_{init}^i \leftarrow$ **ClientUpdate**($i, \mathbf{w}_{init}, F_i(\cdot)$)
    $\Delta \mathbf{w}_{init}^i \leftarrow \mathbf{w}_{init} - \mathbf{w}_{init}^i$
    return $\Delta \mathbf{w}_{init}^i$
**Server Executes**
  $\hat{\mathbf{w}}_{init} = \mathbf{w}_{init} + \sum \frac{n_i}{n} \cdot \Delta \mathbf{w}_{init}^i$
  **for** each client $i \in K$ **do**:
    compute $d_{init}^i = F(\hat{\mathbf{w}}_{init}) - F(\mathbf{w}_{init}^i)$
    return $d_{init}^i$ to each client $i \in K$.

**Algorithm 3: AdaptiveUpdate.** $\mathbf{Win}_i^t$ Is the Window Size for Client $i$ in Round $t$, $E$ Is the Total Number of Epochs, $C$ Is the Clipping Threshold

**AdaptiveUpdate**($i, \mathbf{w}_t, d_t^i$): // *DPSGD run on client i*
  $\mathbf{Win}_i^t \leftarrow$**AdaWin**($temp_i, w_i, t$)
  **for** each local epoch $e$ from $0, ..., E-1$ **do**
    Take a random sample $\mathcal{L}_e$ with sampling probability $q_t^i = \mathcal{L}_t^i / |\mathbf{Win}_i|$.($\mathcal{L}_t^i$: number of samples per epoch)
    **for** each $s \in \mathcal{L}_e$, compute $\mathbf{g}_e(s) \leftarrow \nabla f_i(\mathbf{w}_e; s)$
      $\widetilde{\mathbf{g}}_e(s) \leftarrow \mathbf{g}_e(s) / \max(1, \frac{||\mathbf{g}_e(s)||_2}{C})$
      $\widetilde{\mathbf{g}}_e \leftarrow \frac{1}{\mathcal{L}_t^i}(\sum_{|\mathcal{L}_e|} \widetilde{\mathbf{g}}_e(s) + \mathcal{N}(0, \sigma_{d_t^i}^2 C^2 \mathbf{I}))$
      $\mathbf{w}_{e+1} \leftarrow \mathbf{w}_e - \eta_t^i \widetilde{\mathbf{g}}_e$
      $\Delta \mathbf{w}_{t+1}^i \leftarrow \mathbf{w}_t - \mathbf{w}_E$
  return $\Delta \mathbf{w}_{t+1}^i$ to server.

rate in DPSGD, the self-increasing sample pool will lower the probability of learning new features and delay the evolution of the model. Therefore, we propose to use a sliding window with adaptive size to delimit the range of sampling and training. The initial size is determined by the predefined training set. Each time the client receives a new model, it would adaptively adjust the size according to the addition of new samples. The details are shown in Algorithm 5.

### B. AdaDpFed

This section systematically introduces the algorithm, AdaDpFed. The federated learning architecture is composed

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**Algorithm 4: Client Filtering.** $K_n$ Is Number of Required Clients, $Q$ Is the Threshold for Filtering, $c$ Is the Scale Cofficient, $Q_i$ Represents the Ranking Value for Filtering, $o_i$ Represents the Arriving Order of Results, $S$ Represents the Set Composed by First $K_n$ Returning Results, $S'$ Represents Sorted Version of $S$ in the Descending Order

**Server Executes**
receive the first $K_n$ results as $S$ with order $o_{K_n}$
**for** each client $i \in K_n$:
  compute $Q_i = \frac{1+c/(1+o_i)}{|d_i|}$ and sort $S$
sorted set: $S' \leftarrow S$ (in descending order)
**if** $Q_{|K_n|} > Q$: **return**
**for** each next client $j \in K - K_n$:
  compute $Q_j = \frac{1+c/(1+o_j)}{|d_j|}$
  **if** $Q_j > Q$:
    add $j$ in right index
    **if** $Q_{|K_n|} > Q$: **break**
return $S$.

**Algorithm 5: Adaptive Sliding Window.** There Are $K$ Clients. *temp* Is the Rounds Before the Window Slides, $w$ Is the Scale Cofficient for Adaptive Sliding Window. $n_i^t$ Represents Total Sample Capacity of Client $i$ at Round $t$. $n_{i,t}$ Represents the Starting Index of Chosen Set of Samples

**AdaWin**
**for** each client $i \in K$:
  Set $\mathbf{Win}_{init} \leftarrow n_i^0 = |\mathcal{D}_i|$
**for** each round $t = 0, 1, ..., temp$:
  $\mathbf{Win} \leftarrow \mathbf{Win}_{init}$(training from $n_{i,0}$)
**for** each round $t = temp + 1, temp + 2, ..., T$:
  compute $\Delta n_i = n_i^t - n_i^{t-1}$
  $\mathbf{Win} \leftarrow \mathbf{Win} + \lfloor w \cdot \Delta n_i \rfloor$(training from $n_{i,t}$)
return size **Win**.

of a server and $K$ clients. The number of global aggregation rounds is $T$. The number of training epochs per client is $E$. The evaluating indicator is the highest accuracy after $T$ rounds.

Unlike FedAvg, AdaDpFed's client selection method assumes that the data distribution of each client changes over time. AdaDpFed generates quantification values by calculating the performance distinction between the global model of each round and the local optimal model of each client and then applies them to the adaptive perturbation and client filtering. It is because the updates are highly related to the local dataset which means that local models usually have poor classification ability for unknown classes. This would be reflected in testing performance on the server test set.

At the start of federated training, all clients execute **ClientUpdate**$(\cdot)$ to initialize local models with SGD and return the local minimum. The server then builds the list of the initial quantification values. The absolute value of these values can effectively measure the extent of client drift to adjust the scale of noise addition. Larger noise would be added to clients with a higher degree of non-IID to accelerate the convergence.

In Algorithm 4, the server executes an asynchronous aggregation method to select reliable and powerful clients. It filters out clients with high degrees of non-IID which would slow the convergence in the current round. The threshold is highly correlated with the quantification value. It is different from randomly sampling $K_n$ clients or aggregating the fastest $K_n$ clients. Client filtering records the first $K_n$ updates and then combines them with quantification value and computational speed to generate a value for ranking. After sorting values from highest to lowest, the server compares the last one with the threshold. If it is smaller, the server will continue to receive updates, inserts them into appropriate indexes and drop the last one until the set is larger than or equal to the threshold.

After the set of sampled clients is obtained, the server would stop receiving updates, generate the new model and update the quantification value list broadcast in the next round.

The higher the absolute value of the quantification value means a larger noise scale. Then DPSGD can adaptively calculate the privacy budget by moments accountant as well. The global budget can be calculated by summing all the local budgets. Considering that DPSGD sets a constant ratio for random sampling, the features of new samples may not be learned sufficiently. Moreover, repeatedly training the out-of-date samples is a waste of computational resources and hardly drives the evolution of the model. Therefore, we would like to reduce the probability of sampling old samples and increase the probability of sampling new samples. Algorithm 5 treats all samples stored at clients as an axis and overlays a sliding window to delimit the sampling range. Initially, the size of the window is set as $|n_i|$ for the first *temp* rounds. By adjusting the size of *temp*, Algorithm 5 can control the time point that the window starts to slide. We normally set $temp = T/10$ to ensure enough training epochs for samples over all rounds. As the collection progresses, the window would move on the axis to include more new samples. At the same time, the window would become larger at a slower rate than the sampling speed and the exact speed is determined by the scale coefficient $w$.

The entire federated training would last $T$ rounds in conjunction with the above modules and finally output the global optimal model. The rest of the parameter tuning is elaborated in the experimental section.

## VI. Convergence Analysis

First, we list $L$-smooth, $\mu$-strongly convex, the bounds of the variance of Gaussian noise, and stochastic gradients as four assumptions to be observed throughout the proof.

*Theorem 4 (L-Smooth):* Assuming that for $\forall i \in K$, $F_i$ is $L$-smooth, i.e., for all $\mathbf{x}$ and $\mathbf{y}$, $F_i(\mathbf{x}) \leq F_i(\mathbf{y}) + (\mathbf{x}-\mathbf{y})^T \nabla F_i(\mathbf{y}) + \frac{L}{2}||\mathbf{x}-\mathbf{y}||_2^2$.

*Theorem 5 ($\mu$-Strongly Convex):* Assuming that for $\forall i \in K$, $F_i$ is $\mu$-strongly convex, i.e., for all $\mathbf{x}$ and $\mathbf{y}$, $F_i(\mathbf{x}) \geq F_i(\mathbf{y}) + (\mathbf{x}-\mathbf{y})^T \nabla F_i(\mathbf{y}) + \frac{\mu}{2}||\mathbf{x}-\mathbf{y}||_2^2$.

*Theorem 6:* Let $\sigma^2_{d_t^i}$ be the variance of the Gaussian noise adding to gradients for all $i = 1, \ldots, K$ and $t = 1, \ldots, T-1$. The bounds of $\sigma^2_{d_t^i}$ is bounded as $\mathbb{E}[\sigma^2_{d_t^i}] \leq D^2$.

*Theorem 7:* The bounds of the expected squared norm of stochastic gradients is bounded as $\mathbb{E}||\nabla F_i(\mathbf{w}_t^i, s_t^i)||_2^2 \leq G^2$ for all $i = 1, \ldots, K$ and $t = 1, \ldots, T-1$.

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According to theorems and the convergence of FedAvg on non-IID data [36], we can prove that the proposed method satisfies convergence.

$$\mathbb{E}[F(\overline{\mathbf{w}}_T)] - F(\mathbf{w}^*) \leq \frac{L}{2(T+\beta)} \cdot \left( \frac{\alpha M(T+\beta)(T+\beta-1)}{\alpha\mu(T+\beta)-(T+\beta-1)} + \Delta_1(\beta+1) \right) \tag{4}$$

where $M = \eta_{T-1}E^2D^2C^2 + (2 + 2\eta_{T-1}^2 L)\Lambda + (1 + 8\mathcal{T})E^2\eta_{T-1}\max\{C, G\}^2$, $\Lambda = F_i(\mathbf{w}^*) - \sum_{i=1}^K p_i F_i^*$, $F^*$ and $F_i^*$ are the minimum values of $F$ and $F_i$, $\eta_t = \frac{\alpha}{t+\beta}$.

*Proof:* From Theorem 3, it follows that

$$\mathbb{E}[F(\overline{\mathbf{w}}_T)] - F(\mathbf{w}^*) \leq \frac{L}{2} \cdot \mathbb{E}||\overline{\mathbf{w}}_T - \mathbf{w}^*||_2^2 \tag{5}$$

where $\overline{\mathbf{w}}_T = \sum_{i=1}^K p_i \mathbf{w}_T^i$ and $p_i = \frac{n_i}{n}$. $\mathbf{w}_T^i$ represents the returned calculation results of client $i \in K$ at the final round $T$. The other returned results at other rounds is represented as $\mathbf{w}_t^i$ where $t \in [T]$. When each client train $E$ epochs at each round, $\mathbf{w}_{t+1}^i = \overline{\mathbf{w}}_t - E\eta_t \widetilde{\mathbf{g}}_t^i$, where $\widetilde{\mathbf{g}}_t^i = \overline{\mathbf{g}}_t^i + \mathcal{N}(0, \sigma_{d_t^i}^2 C^2 \mathbf{I})$ and $\overline{\mathbf{g}}_t^i = \frac{1}{\mathcal{L}} \sum_{l \in \mathcal{L}} \mathbf{g}_t^i(x_l) / \max\left(1, \frac{||\mathbf{g}_t^i(x_l)||}{C}\right)$. Therefore,

$$\begin{aligned} \overline{\mathbf{w}}_{t+1} &= \sum_{i=1}^K p_i \mathbf{w}_{t+1}^i = \overline{\mathbf{w}}_t - \sum_{i=1}^K p_i E\eta_t \widetilde{\mathbf{g}}_t^i \\ &= \overline{\mathbf{w}}_t - \eta_t \left( \sum_{i=1}^K p_i E \overline{\mathbf{g}}_t^i \right) - \eta_t \left( \sum_{i=1}^K p_i E \mathcal{N}\left(0, \sigma_{d_t^i}^2 C^2 \mathbf{I}\right) \right) \end{aligned} \tag{6}$$

Let $\hat{\mathbf{g}}_t = \sum_{i=1}^K p_i E \overline{\mathbf{g}}_t^i$ and $\hat{\mathcal{N}}_t = \sum_{i=1}^K p_i E \mathcal{N}(0, \sigma_{d_t^i}^2 C^2 \mathbf{I})$, and $\hat{\mathbf{g}}_t$ can be rewriten as $\overline{\mathbf{w}}_{t+1} = \overline{\mathbf{w}}_t - \eta_t \hat{\mathbf{g}}_t - \eta_t \hat{\mathcal{N}}_t$. Then we transform $||\overline{\mathbf{w}}_T - \mathbf{w}^*||_2^2$ into

$$\begin{aligned} &||\overline{\mathbf{w}}_{T-1} - \mathbf{w}^*||_2^2 - \underbrace{2\eta_{T-1}\langle \overline{\mathbf{w}}_{T-1} - \mathbf{w}^*, \widetilde{\mathbf{g}}_{T-1}^i \rangle}_{A_1} \\ &+ \underbrace{\eta_{T-1}^2 ||\hat{\mathbf{g}}_{T-1} + \hat{\mathcal{N}}_{T-1}||_2^2}_{A_2} \end{aligned} \tag{7}$$

Next, we calculate $\mathbb{E}A_1$ and $\mathbb{E}A_2$ respectively.

$$\begin{aligned} \mathbb{E}A_1 &= 2\eta_{T-1}\mathbb{E}\left\langle \overline{\mathbf{w}}_{T-1} - \sum_{i=1}^K p_i \mathbf{w}_{T-1}^i + \sum_{i=1}^K p_i \mathbf{w}_{T-1}^i - \mathbf{w}^*, \widetilde{\mathbf{g}}_{T-1}^i \right\rangle \\ &= 2\eta_{T-1}\left\{ \mathbb{E}\left[ \sum_{i=1}^K p_i \langle \overline{\mathbf{w}}_{T-1} - \mathbf{w}_{T-1}^i, \nabla F_i(\mathbf{w}_{T-1}^i) \rangle \right] \right. \\ &\left. + \mathbb{E}\left[ \sum_{i=1}^K p_i \langle \mathbf{w}_{T-1}^i - \mathbf{w}^*, \nabla F_i(\mathbf{w}_{T-1}^i) \rangle \right] \right\} \end{aligned} \tag{8}$$

By Cauchy-Schwarz inequality, Inequality of Arithmetic and Geometric Means (AM-GM inequality) and the $\mu$-strongly convex of $F_{T-1}^i(\cdot)$, we have the following two inequalities,

$$\begin{aligned} &-2\langle \overline{\mathbf{w}}_{T-1} - \mathbf{w}_{T-1}^i, \nabla F_i(\mathbf{w}_{T-1}^i) \rangle \leq \frac{1}{\eta_{T-1}} ||\overline{\mathbf{w}}_{T-1} - \mathbf{w}_{T-1}^i||_2^2 \\ &+ \eta_{T-1} ||\nabla F_i(\mathbf{w}_{T-1}^i)||_2^2 \end{aligned} \tag{9}$$

$$\begin{aligned} &-\langle \mathbf{w}_{T-1}^i - \mathbf{w}^*, \nabla F_i(\mathbf{w}_{T-1}^i) \rangle \leq -\left(F_i(\mathbf{w}_{T-1}^i) - F_i(\mathbf{w}^*)\right) \\ &- \frac{\mu}{2} ||\mathbf{w}_{T-1}^i - \mathbf{w}^*||_2^2 \end{aligned} \tag{10}$$

Therefore,

$$\begin{aligned} -A_1 &\leq \eta_{T-1} \sum_{i=1}^K p_i \left( \frac{1}{\eta_{T-1}} ||\overline{\mathbf{w}}_{T-1} - \mathbf{w}_{T-1}^i||_2^2 + \eta_{T-1} \right. \\ &\left. ||\nabla F_i(\mathbf{w}_{T-1}^i)||_2^2 \right) - 2\eta_{T-1} \sum_{i=1}^K p_i \left( \left(F_i(\mathbf{w}_{T-1}^i) - F_i(\mathbf{w}^*)\right) \right. \\ &\left. - \frac{\mu}{2} ||\mathbf{w}_{T-1}^i + \mathbf{w}^*||_2^2 \right) \\ &= p_i \eta_{T-1}^2 ||\nabla F_i(\mathbf{w}_{T-1}^i)||_2^2 + \sum_{i=1}^K \left( p_i ||\overline{\mathbf{w}}_{T-1} - \mathbf{w}_{T-1}^i||_2^2 \right) \\ &- 2\eta_{T-1} \sum_{i=1}^K p_i \left(F_i(\mathbf{w}_{T-1}^i) - F_i(\mathbf{w}^*)\right) - \mu\eta_{T-1} \\ &\times \sum_{i=1}^K p_i ||\mathbf{w}_{T-1}^i - \mathbf{w}^*||_2^2 \end{aligned} \tag{11}$$

For $||\nabla F_i(\mathbf{w}_{T-1}^i)||_2^2 \leq 2L(F_i(\mathbf{w}_{T-1}^i) - F_i^*)$,

$$\begin{aligned} -A_1 &\leq -\mu\eta_{T-1} ||\overline{\mathbf{w}}_{T-1} - \mathbf{w}^*||_2^2 + \sum_{i=1}^K p_i ||\overline{\mathbf{w}}_{T-1} - \mathbf{w}_{T-1}^i||_2^2 \\ &+ 2\eta_{T-1} \left( \eta_{T-1} L \sum_{i=1}^K p_i \left(F_i(\mathbf{w}_{T-1}^i) - F_i^*\right) \right. \\ &\left. - \sum_{i=1}^K p_i \left(F_i(\mathbf{w}_{T-1}^i) - F_i(\mathbf{w}^*)\right) \right) \end{aligned} \tag{12}$$

Then, we compute the last three terms respectively. First, by AM-GM inequality and Theorem 3, we have

$$\begin{aligned} &2\eta_{T-1} \left( \eta_{T-1} L \sum_{i=1}^K p_i \left(F_i\left(\mathbf{w}_{T-1}^i\right) - F_i^*\right) - \sum_{i=1}^K p_i \left(F_i\left(\mathbf{w}_{T-1}^i\right)\right) \right. \\ &\left. - F_i(\mathbf{w}^*) \right) \\ &= -\left(1 - \eta_{T-1}\right)^2 L \sum_{i=1}^K p_i \left(F_i(\overline{\mathbf{w}}_{T-1}) - F^*\right) \\ &+ \left(1 - \eta_{T-1} L\right) \sum_{i=1}^K p_i ||\mathbf{w}_{T-1}^i - \overline{\mathbf{w}}_{T-1}||_2^2 \\ &+ \left(2 + 2\eta_{T-1}^2 (1 - \eta_{T-1} L) L\right) \eta_{T-1} \left( F_i(\mathbf{w}^*) - \sum_{i=1}^K p_i F_i^* \right) \\ &\leq \sum_{i=1}^K p_i ||\mathbf{w}_{T-1}^i - \overline{\mathbf{w}}_{T-1}||_2^2 \\ &- \eta_{T-1} \left( \sum_{i=1}^K p_i F_i^* - F_i(\mathbf{w}^*) \right) \left(2 + 2\eta_{T-1}^2 L\right) \end{aligned} \tag{13}$$

Secondly, we use the fact that $\eta_t$ is non-increasing and $T - t_0 \leq \mathcal{T}$,

$$\begin{aligned} &\sum_{i=1}^K p_i ||\overline{\mathbf{w}}_{T-1} - \mathbf{w}_{T-1}^i||_2^2 \leq \sum_{i=1}^K p_i ||\mathbf{w}_{t_0}^i - \overline{\mathbf{w}}_{T-1}||_2^2 \\ &\leq \sum_{i=1}^K p_i \mathcal{T} E^2 \eta_{t_0}^2 ||\overline{\mathbf{g}}_t^i||^2 = \mathcal{T} E^2 \eta_{t_0}^2 \max\{C, G\}^2 \end{aligned} \tag{14}$$

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Therefore,

$$-A_1 \le -\mu\eta_{T-1}||\overline{\mathbf{w}}_{T-1}-\mathbf{w}^*||_2^2 + 2\mathcal{T}E^2\eta_0^2\max\{C,G\}^2 + \Big(2+2\eta_{T-1}^2L\Big)\eta_{T-1}\left(F_i(\mathbf{w}^*)-\sum_{i=1}^{K}p_iF_i^*\right) \quad (15)$$

Also,

$$\mathbb{E}A_2 = \eta_{T-1}^2\mathbb{E}||\hat{\mathbf{g}}_{T-1}+\hat{\mathcal{N}}_{T-1}||_2^2 \le \eta_{T-1}^2\mathbb{E}\Big[\hat{\mathbf{g}}_{T-1}^2\Big] + \eta_{T-1}^2\mathbb{E}\Big[\hat{\mathcal{N}}_{T-1}^2\Big] + 2\eta_{T-1}^2\mathbb{E}\Big[\hat{\mathbf{g}}_t\hat{\mathcal{N}}_{T-1}\Big] \quad (16)$$

By Theorem 4 and the variance of Gaussian distribution, we have $\eta_{T-1}^2\mathbb{E}[\hat{\mathbf{g}}_{T-1}^2] \le \eta_{T-1}^2E^2\max\{C,G\}^2$ and $\eta_{T-1}^2\mathbb{E}[\hat{\mathcal{N}}_{T-1}^2] \le \eta_{T-1}^2E^2\sum_{i=1}^{K}(p_i\sigma_{d_{T-1}^i})^2C^2$. For $\mathbb{E}[\hat{\mathbf{g}}_t\hat{\mathcal{N}}_{T-1}]=0$, we have

$$\mathbb{E}A_2 \le \eta_{T-1}^2E^2\max\{C,G\}^2 + \eta_{T-1}^2E^2\sum_{i=1}^{K}\Big(p_i\sigma_{d_{T-1}^i}\Big)^2C^2. \quad (17)$$

For simplicity of expression, we use $\Lambda = F_i(\mathbf{w}^*)-\sum_{i=1}^{K}p_iF_i^*$. To sum up,

$$\begin{aligned}\mathbb{E}||\overline{\mathbf{w}}_T-\mathbf{w}^*||_2^2 &\le \mathbb{E}||\overline{\mathbf{w}}_{T-1}-\mathbf{w}^*||_2^2 - \mathbb{E}A_1 + \mathbb{E}A_2\\ &= (1-\mu\eta_{T-1})\mathbb{E}||\overline{\mathbf{w}}_{T-1}-\mathbf{w}^*||_2^2 + \Big(2+2\eta_{T-1}^2L\Big)\eta_{T-1}\Lambda\\ &\quad + \eta_{T-1}^2E^2\max\{C,G\}^2 + 2\mathcal{T}E^2\eta_0^2\max\{C,G\}^2\\ &\quad + \eta_{T-1}^2E^2\sum_{i=1}^{K}p_i^2D^2C^2 \end{aligned} \quad (18)$$

For further simplicity, we let $\Delta_T = \mathbb{E}||\overline{\mathbf{w}}_T-\mathbf{w}^*||_2^2$. Then we use the following facts: (1) $\eta_0 < 2\eta_{T-1}$, (2) $\sum_{i=1}^{K}p_i^2 \le 1$, then the above formula can be rewritten as

$$\Delta_T \le (1-\mu\eta_{T-1})\Delta_{T-1} + \eta_{T-1}^2E^2D^2C^2 + \Big(2+2\eta_{T-1}^2L\Big)\eta_{T-1}\Lambda + (1+8\mathcal{T})E^2\eta_{T-1}^2\max\{C,G\}^2 \quad (19)$$

Let $M = \eta_{T-1}E^2D^2C^2 + (2+2\eta_{T-1}^2L)\Lambda + (1+8\mathcal{T})E^2\eta_{T-1}\max\{C,G\}^2$, then

$$\Delta_T \le (1-\mu\eta_{T-1})\Delta_{T-1} + \eta_{T-1}M. \quad (20)$$

For a monotonically decreasing learning rate, we assume that $\eta_{t-1} = \frac{\alpha}{t-1+\beta}$ where $t=1,\dots,T-1$, $\alpha>0$, $\beta>0$. By induction, we can prove $\Delta_T \le \frac{v}{T+\beta}$.

$$\begin{aligned}\Delta_T &\le (1-\mu\eta_{T-1})\Delta_{T-1} + \eta_{T-1}M\\ &= \left(1-\frac{\mu\alpha}{T-1+\beta}\right)\Delta_{T-1} + \frac{\alpha M}{T-1+\beta}\\ &\le \left(1-\frac{\mu\alpha}{T-1+\beta}\right)\frac{v}{T-1+\beta} + \frac{\alpha M}{T-1+\beta}\\ &= \frac{v}{T+\beta} + \frac{1}{T-1+\beta}\left(\frac{(T+\beta-1)-\alpha\mu(T+\beta)}{(T+\beta)(T+\beta-1)}v + \alpha M\right) \le \frac{v}{T+\beta}\end{aligned} \quad (21)$$

where $v = \max\left\{\frac{\alpha M(T+\beta)(T+\beta-1)}{\alpha\mu(T+\beta)-(T+\beta-1)}, \Delta_1(\beta+1)\right\}$. Finally, by Theorem 3 we have

$$\begin{aligned}\mathbb{E}\big[F(\overline{\mathbf{w}})_T\big] - F(\mathbf{w}^*) &\le \frac{L}{2}\frac{v}{T+\beta}\\ &= \frac{L}{2(T+\beta)}\cdot\max\left\{\frac{\alpha M(T+\beta)(T+\beta-1)}{\alpha\mu(T+\beta)-(T+\beta-1)}, \Delta_1(\beta+1)\right\}\\ &\le \frac{L}{2(T+\beta)}\cdot\left(\frac{\alpha M(T+\beta)(T+\beta-1)}{\alpha\mu(T+\beta)-(T+\beta-1)} + \Delta_1(\beta+1)\right)\end{aligned} \quad (22)$$

■

## VII. Empirical Results

Experiment evaluates performance on the MNIST handwritten digit recognition task. The baseline takes FedAvg+DPSGD, DP-FedAvg and FL-PATE. Then we compare AdaDpFed in different scales of noise distribution, different thresholds for client filtering, and different adaptive window sizes.

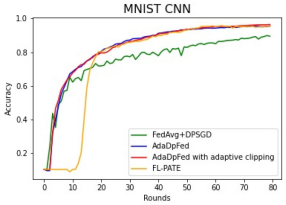
*Data Partition:* The training datasets are personalized for clients to ensure the non-IID setting. We reallocate the data as follows. An upper bound of the number of classes is randomly selected for each client. Even if two clients' upper bounds are the same, the specific classes would still be different. Another consideration is the self-increment of client samples. Since the experiment simulates real-world sample collection, the total number of samples per client would grow in each round. The new samples can be arbitrary classes and thus the degree of non-IID would change over the data distribution. For this reason, the experiment divides 60,000 training samples into the initial datasets and supplementary datasets. The test set includes 10,000 samples in total, and all the clients are sharing the same test set.

*Baseline:* The baseline takes FedAvg with DPSGD. Meanwhile, some perturbation algorithms in other DP-based federated learning protocols, such as DP-FedAvg and FL-PATE, have been referenced in the comparative experiments. Specifically, we respectively add an adaptive clipping module from DP-FedAvg [30] into AdaDpFed and extract the FL part of FL-PATE [31].

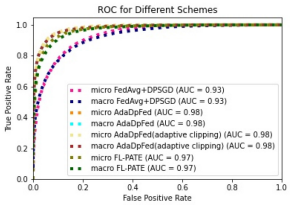
The experiments set up 100 clients that adopt the same data partition in all schemes. All clients have an initial state and initial dataset. We set the same training parameters for both the server and clients. Except for AdaDpFed with adaptive clipping, the gradient clipping parameter $C$ is fixed for all schemes. In the comparative group, we run 80 rounds and choose 40 clients to be aggregated at each round. The results are followed in Fig. 4(a) and recorded in Table I.

The performance of AdaDpFed exceeds the FedAvg + DPSGD at round 4, and it keeps widening the gap. The accuracy of AdaDpFed eventually comes to 95.85%, while FedAvg+DPSGD comes to 89.48%. When adding an adaptive clipping module into AdaDpFed, there is only a 0.6% improvement of final accuracy relying on an additional budget of 0.013. Meanwhile, since FL-PATE depends on layer-based perturbation, it would consume more budget to achieve the same final accuracy. To achieve a final accuracy of 95%,

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(a) Federated Learning




(b) ROC for FedAvg+DPSGD vs AdaDpFed

Fig. 4. The accuracy and ROC of different protocols.

FL-PATE requires 8.5× budget of AdaDpFed. If a budget similar to other protocols is adopted, FL-PATE would have almost no performance improvement. In general, AdaDpFed has better privacy than FedAvg+DPSGD and FL-PATE and has no indispensable requirement for adaptive clipping.

To comprehensively compare the impact of protocols on model prediction performance, we draw ROC curves to verify the classification accuracy and use AUC to evaluate the quality of classifiers. ROC is usually used to evaluate the predictive ability of models, but mostly binary classification models. Therefore, we use two reasonable approaches to make it feasible for evaluating multi-classification models. (a) We take the average of 10 ROC curves of each class to get the final ROC curve. (b) We compare the predicted probability of each sample with its binarized label to directly get the ROC curve. The results are followed in Fig. 4(b). The micro-average ROC curve and macro-average ROC curve correspond to approach (a) and approach (b) respectively. The results show that the AUCs of AdaDpFed are both 0.98 no matter whether introducing the adaptive clipping module while the counterparts of FedAvg+DPSGD and FL-PATE are 0.93 and 0.97, which means AdaDpFed trains a better classifier in terms of classification accuracy in the non-IID setting.

*Parameter Tuning:* According to the features of the proposed methods in AdaDpFed, we conduct a series of comparative experiments to adjust the relevant parameters and record the results in Table I. The specific parameters involved in the experiments are the number of clients filtered for each round of aggregation ($K$), the number of epochs that the client performs local training ($E$), the average budget required by clients per round ($\overline{\epsilon}$), the threshold for client filtering ($Q$), and the initial size of adaptive sliding window ($\Delta\mathbf{Win}_{init}$). The $\overline{\epsilon}$ is adaptively calculated by moments accountant proposed by Abadi et al. [4] and represents the average budget over $n$ clients. In Tensorflow-privacy, DPSGD has been integrated with Tensorflow and uses the ratio of standard deviation to the clipping norm to modulate noise. Therefore, according to the client's quantification value of each round, we can output the $\overline{\epsilon}$ correspondingly.

### A. Adaptive Perturbation

AdaDpFed adaptively increases the scale of noise for clients that have a higher degree of non-IID to improve the utility of local models and help the global model speed up the convergence. When the required average client budget is counted for each round, AdaDpFed would consume less budget than the baseline. If we set the total budget value to a fixed value, the experimental accuracy might be further improved.

TABLE I
ACCURACY OVER 80 ROUNDS FOR THE MNIST HANDWRITTEN DIGIT RECOGNITION TASK IN THE NON-IID SETTING

| Protocol | $K$ | $E$ | $\overline{\epsilon}$ | $Q$ | $\triangle\mathbf{Win}_{init}$ | Accuracy |
|---|---|---|---|---|---|---|
| FedAvg+DPSGD | 40 | 5 | 2.78 | – | – | 89.48% |
| DP-FedAvg | 40 | 5 | 2.70 | 4 | 64 | 96.45% |
| FL-PATE | 40 | 5 | 22.8 | 4 | 64 | 95.30% |
| AdaDpFed | 10 | 5 | 2.69 | 4 | 64 | 94.12% |
| AdaDpFed | 40 | 5 | 2.68 | 4 | 64 | 95.85% |
| AdaDpFed | 20 | 5 | 2.69 | 4 | 64 | 95.38% |
| AdaDpFed | 30 | 5 | 2.68 | 4 | 64 | 95.62% |
| AdaDpFed | 40 | 1 | 1.54 | 4 | 64 | 83.98% |
| AdaDpFed | 40 | 2 | 1.88 | 4 | 64 | 91.05% |
| AdaDpFed | 40 | 10 | 3.68 | 4 | 64 | 96.88% |
| AdaDpFed | 40 | 5 | 2.68 | 4 | 32 | 93.94% |
| AdaDpFed | 40 | 5 | 2.68 | 4 | 48 | 95.41% |
| AdaDpFed | 40 | 5 | 2.66 | 4 | 80 | 96.35% |
| AdaDpFed | 40 | 5 | 2.67 | 1 | 64 | 95.28% |
| AdaDpFed | 40 | 5 | 2.67 | 2 | 64 | 95.44% |
| AdaDpFed | 40 | 5 | 2.68 | 8 | 64 | 96.21% |

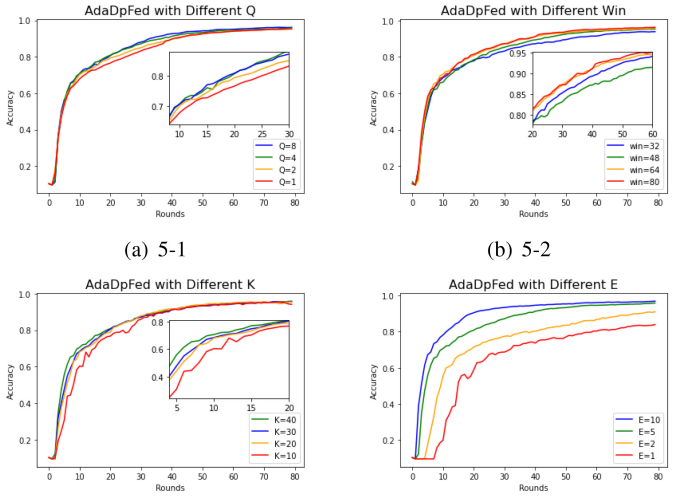
### B. Optimal Threshold Q

The threshold $Q$ is one of the most crucial parameters which is closely related to the quantification value and the communication efficiency. Algorithm 4 integrates and deflates them to compute a $Q_i$ for ranking all the clients. The ranking list of the individual clients is changing over the varying data distribution, thus every client might be a potential aggregated party. Based on the sorted $Q_i$ values, we set a threshold to filter the clients. The threshold has a nice feature that there is a superior bound on its optimality. For example, the accuracy curve is almost the same as $Q = 4$ and $Q = 8$, which means that the two thresholds have similar effects. When $Q$ is further reduced, the corresponding curves are clearly distinguished from the first two. This indicates that there is an optimal value of $Q$. Benefiting from this, the training would reach the target accuracy with a low time cost.

### C. Different Initial Window Sizes

In industrialized scenarios, the increase of the size of sliding window is decided by the growing samples of each round. Ensuring that each sample is fully learned, the window would grow with a predefined scale coefficient $w$. It can be seen that as $\Delta\mathbf{Win}_{init}$ increases, the accuracy improves but the time cost is much smaller than enlarging the number of local epochs $E$ or the total number of training rounds $T$. This is because the method incorporates learning new features while fully training the old samples. However, this increase in accuracy is not strictly linear. In Fig. 5-2, there is a more pronounced distinction between the corresponding accuracy curves when $\Delta\mathbf{Win}_{init}$ is at 32, 48, and 64, but when the window is further enlarged, the accuracy is almost no longer increased. This illustrates that there is an optimal choice of $\Delta\mathbf{Win}_{init}$ for a specific task.

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(a) 5-1 (b) 5-2

(c) 5-3 (d) 5-4

Fig. 5. AdaDpFed with different parameters.

### D. Different Aggregation Numbers

The aggregation number has a different meaning in the baseline and AdaDpFed. In baseline, it represents $K$ random updates for each round. In AdaDpFed, it represents the first $K$ updates satisfying the threshold. The accuracy is highly related to the size of $K$. Initially, the accuracy is improved steadily as $K$ increases and then the impact would be gradually reduced. It is well illustrated in Fig. 5-3 that there is a clear but gradually decreasing gap between the accuracy curves. Between $K = 30$ and $K = 40$, the gap has narrowed to a very close extent. This number represents the portion of clients, which have a lower degree of non-IID and a more reasonable distribution in a non-IID environment which can be considered as a stable number. A greater number may be sufficient to achieve the best accuracy but it would simultaneously cost more time. Conversely, it would lead to a slow convergence rate. Therefore, by adjusting the value of $K$, the algorithm can accelerate the process for achieving fast convergence.

### E. The Impact of E

When adjusting the number of epochs per round for each local client, a simple intuitive is that a larger number of local epochs can lead to higher local training accuracy. Naturally, the training time per round linearly grows as well. However, the training accuracy does not grow linearly. Therefore, it is not worth spending a lot of time on small accuracy improvements. As can be seen in Table I, the final accuracy improves only 1.03% when $E$ is raised from 5 to 10 but takes twice the time. Conversely, the improvement in accuracy is more obvious when $E$ is at smaller values, such as $E = 1$ and $E = 2$. Meanwhile, it can be seen that the model can obtain relatively good convergence at $E = 5$. Compared with the original baseline with $E = 5$, AdaDpFed can set $E = 2$ to achieve better model accuracy, which undoubtedly saves time cost.

## VIII. Conclusion

In this paper, we proposed AdaDpFed, an adaptive privacy-preserving federated learning protocol for non-IID environments. AdaDpFed guarantees flexible user-level differential privacy of each client to defend against external attacks. Within the protocol, a client filtering algorithm and adaptive sampling algorithm are proposed to accelerate the convergence in the non-IID environment and reduce the impact of the perturbation. The theoretical guarantees and empirical results indicate that AdaDpFed outperforms other state-of-the-art protocols in both privacy and accuracy. Future work should further explore more potential possibilities related to adaptability, such as the combination of privacy budget, hyperparameters and updating strategies. Some other security technologies can also be introduced to increase the privacy level.

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